

1) Fraud Transaction detection (linear search)

1201 1305 1450 1408 1502
i = 0 1 2 3 4

find fraud id.

Code

```
int arr[] = {1201, 1305, 1450, 1408, 1502}
```

```
S.O.P ("Enter Fraud Id");
```

```
int FId = Sc. nextInt();
```

```
boolean found = false;
```

```
for (int i = 0; i < arr.length; i++) {  
    if (arr[i] == FId) {  
        found = true;  
        break;  
    }  
}
```

```
if (found) {  
    S.O.P ("Found");
```

```
else  
    S.O.P ("Not Found");
```

```
}
```



2) Linear log search (Binary Search)

	1001	1005	1010	1015	1020	1030	1040
i =	↑ 0	1	2	3 ↑	4	5	6 ↑
	low			mid			high

while (low <= high)

mid = low + $\frac{(high - low)}{2}$

Code

```
int low = 0;
```

```
int high = arr.length - 1;
```

```
while (low <= high) {
```

```
    int mid = low + (high - low) / 2;
```

```
    if (arr[mid] == target) {
```

```
        S.O.P("Element Found");
```

```
    }
```

```
    else if (arr[mid] < target) {
```

```
        low = mid + 1;
```

```
    }
```

```
    else {
```

```
        high = mid - 1;
```

```
    }
```

```
}
```

```
S.O.P("Not Found");
```


3] Sort Employee Salaries (Bubble sort)

Pass 1	j=	45000	32000	55000	28000	60000
		arr[j]	arr[j+1]			
		swap				
		$arr[j] > arr[j+1]$				
	j=2	32000	45000	55000	28000	60000
		no swap				
	j=3	32000	55000	45000	28000	60000
		swap				
	j=4	32000	55000	28000	45000	(60000) → fixed
	j=5	32000	55000	28000	45000	(60000)
Pass 2	j=	32000	55000	28000	45000	(60000)
		do not swap				
	j=2	32000	55000	28000	45000	(60000)
		swap				
	j=3	32000	28000	55000	45000	(60000)
		swap				
	j=4	32000	28000	45000	(55000)	(60000)
Pass 3	j=	32000	28000	45000	(55000)	(60000)
		swap				
	j=2	28000	32000	45000	(55000)	(60000)
		do not swap				
	j=3	28000	32000	(45000)	(55000)	(60000)
Pass 4	j=	28000	32000	(45000)	(55000)	(60000)
		do not swap				
	j=2	(28000)	(32000)	(45000)	(55000)	(60000)

```
for (i = 0; i < N; i++) {  
    for (j = 0; j < N - i - 1; j++) {  
        if (arr[j] > arr[j + 1]) {  
            int temp = arr[j];  
            arr[j] = arr[j + 1];  
            arr[j + 1] = temp;  
        }  
    }  
}
```

}

}

}

47) Sort website response time (Insertion Sort)

250 120 320 180 100
Compare j & $j++$:- check for elements &
insert it at correct place

(250) | 120 320 180 100
↖

(120) (250) | 320 180 100
↖

(120) (250) (320) | 180 100
↖

(120) (180) (250) (320) | 100
↖

(100) (120) (180) (250) (320) (sorted)

```
for (int i = 1; i < N; i++) {  
    int key = arr[i];  
    int j = i - 1;
```

```
    while (j >= 0 && arr[j] > key) {  
        arr[j+1] = arr[j];  
        j--;
```

```
    }  
    arr[j+1] = key;
```

```
}
```

Print sorted array

5) Bug Reports (Selection sort)

9 2 8 1 5

Logic

at every step/pass, find smallest element and put it into first.

Pass ①

9 2 8 ① 5

= 1 9 2 8 5

Pass ②

2 is smallest

2 is at correct position

Pass ③

1 2 : 8 9 ⑤

1 2 5 : 8 9

Pass ④

Now, 8 is smallest

It is at correct position

Code :-

```
for (int i = 0; i < arr.length - 1; i++) {
```

```
    int min = i;
```

```
    for (int j = i + 1; j < arr.length; j++) {
```

```
if (arr[j] < arr[min]) {  
    min = j ;
```

```
}
```

```
int temp = arr[i];  
arr[i] = arr[min];  
arr[min] = temp;
```

```
}
```

```
for (int i = 0 ; i < arr.length ; i++) {  
    S.O.P (arr[i] + " ");
```

```
}
```

```
}
```


⚡

Main.java

Run

Debug

Stop

Share

Save

{}

Beautify

Language

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         //Linear Search
8         int arr[] = {1201, 1305, 1450, 1408, 1502};
9         System.out.print("Enter Fraud ID: ");
10        int fId = sc.nextInt();
11
12        boolean found = false;
13        for (int i = 0; i < arr.length; i++) {
14            if (arr[i] == fId) {
15                found = true;
16                break;
17            }
18        }
19        if (found) {
20            System.out.println("Fraud Transaction Found");
21        } else {
22            System.out.println("Not Found");
23        }
24    }
25 }
```

input

Enter Fraud ID: 1450
Fraud Transaction Found

...Program finished with exit code 0
Press ENTER to exit console.

3

Search

08:13 PM
22-06-2026

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Main.java ⋮

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         // Binary Search
8         int arr[] = {1001, 1005, 1010, 1015, 1020, 1030, 1040};
9         System.out.println("Enter no. to be checked");
10        int target = sc.nextInt();
11        int low = 0;
12        int high = arr.length - 1;
13        while (low <= high) {
14            int mid = low + (high - low) / 2;
15            if(arr[mid] == target) {
16                System.out.println("Element Found");
17                return; }
18            else if(arr[mid] < target) {
19                low = mid + 1; }
20            else {
21                high = mid - 1; }
22        }
23        System.out.println("Not Found");
24    }
25 }

```

Enter no. to be checked

1010

Element	Found
Hydrogen	1
Helium	2
Lithium	3
Boron	5
Carbon	6
Nitrogen	7
Oxygen	8
Fluorine	9
Neon	10
Sodium	11
Magnesium	12
Aluminum	13
Silicon	14
Phosphorus	15
Sulfur	16
Chlorine	17
Argon	18
Potassium	19
Calcium	20
Scandium	21
Titanium	22
Vanadium	23
Chromium	24
Manganese	25
Iron	26
Cobalt	27
Nickel	28
Copper	29
Zinc	30
Gallium	31
Germanium	32
Arsenic	33
Selenium	34
Bromine	35
Krypton	36
Rubidium	37
Strontium	38
Yttrium	39
Zirconium	40
Niobium	41
Molybdenum	42
Technetium	43
Ruthenium	44
Rhodium	45
Palladium	46
Silver	47
Cadmium	48
Indium	49
Sn	50
Antimony	51
Tellurium	52
Bismuth	53
Polonium	54
Astatine	55
Radon	56
Francium	57
Radium	58
Actinium	59
Thorium	60
Protactinium	61
Uranium	62
Neptunium	63
Plutonium	64
Americium	65
Cerium	66
Praseodymium	67
Neodymium	68
Promethium	69
Samarium	70
Europium	71
Gadolinium	72
Terbium	73
Dysprosium	74
Ho	75
Erbium	76
Thulium	77
Ytterbium	78
Lanthanum	79
Ce	80
Pr	81
Nd	82
Pm	83
Sm	84
Eu	85
Gd	86
Tb	87
Dy	88
Ho	89
Er	90
Tm	91
Yb	92
Lu	93
Hafnium	94
Tantalum	95
Tungsten	96
Rhenium	97
Osmium	98
Iridium	99
Pt	100
Au	101
Hg	102
Thallium	103
Lead	104
Bismuth	105
Polonium	106
Astatine	107
Radon	108
Francium	109
Radium	110
Actinium	111
Thorium	112
Protactinium	113
Uranium	114
Neptunium	115
Plutonium	116
Americium	117
Cerium	118
Praseodymium	119
Neodymium	120
Promethium	121
Samarium	122
Europium	123
Gadolinium	124
Terbium	125
Dysprosium	126
Ho	127
Erbium	128
Thulium	129
Ytterbium	130
Lanthanum	131
Ce	132
Pr	133
Nd	134
Pm	135
Sm	136
Eu	137
Gd	138
Tb	139
Dy	140
Ho	141
Er	142
Tm	143
Yb	144
Lu	145
Hafnium	146
Tantalum	147
Tungsten	148
Rhenium	149
Osmium	150
Iridium	151
Pt	152
Au	153
Hg	154
Thallium	155
Lead	156
Bismuth	157
Polonium	158
Astatine	159
Radon	160
Francium	161
Radium	162
Actinium	163
Thorium	164
Protactinium	165
Uranium	166
Neptunium	167
Plutonium	168
Americium	169
Cerium	170
Praseodymium	171
Neodymium	172
Promethium	173
Samarium	174

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.
```

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Run

Debug

Stop

Share

Save

Beautify

Language

Main.java

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int arr[] = new int[n];
7         for(int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        for(int i = 0; i < n-1; i++) {
11            for(int j = 0; j < n-1-i; j++) {
12                if(arr[j] > arr[j+1]) {
13                    int temp = arr[j];
14                    arr[j] = arr[j+1];
15                    arr[j+1] = temp;
16                } } }
17        for(int i = 0; i < n; i++) {
18            System.out.print(arr[i] + " ");
19        } }
```

input

5
45000
32000
55000
28000
60000
28000 32000 45000 55000 60000

...Program finished with exit code 0
Press ENTER to exit console.

31°C Mostly cloudy

Search

3

ENG IN

08:08 PM 22-06-2026

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Run Debug Stop Share Save Beautify Language input

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int N = sc.nextInt();
7         int arr[] = new int[N];
8
9         for(int i = 0; i < N; i++) {
10             arr[i] = sc.nextInt();
11         }
12         for(int i = 1; i < N; i++) {
13             int key = arr[i];
14             int j = i - 1;
15
16             while(j >= 0 && arr[j] > key) {
17                 arr[j + 1] = arr[j];
18                 j--;
19             }
20             arr[j + 1] = key;
21         }
22         for(int i = 0; i < N; i++) {
23             System.out.print(arr[i] + " ");
24         }
25     }
26 }
```

5
250
120
320
180
100
100 120 180 250 320

...Program finished with exit code 0
Press ENTER to exit console.

3 Search ENG IN 08:10 PM 22-06-2026





Language



input

Main.java

```
1 import java.util.Scanner;
2
3 public class Main{
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         int N = sc.nextInt();
9         int arr[] = new int[N];
10        for(int i = 0; i < N; i++) {
11            arr[i] = sc.nextInt();
12        }
13        for(int i = 0; i < N - 1; i++) {
14            int min = i;
15            for(int j = i + 1; j < N; j++) {
16                if(arr[j] < arr[min]) {
17                    min = j;
18                }
19            }
20            int temp = arr[i];
21            arr[i] = arr[min];
22            arr[min] = temp;
23        }
24        for(int i = 0; i < N; i++) {
25            System.out.print(arr[i] + " ");
26        }
27    }
28 }
```

```
5
9
2
8
1
5
1 2 5 8 9

...Program finished with exit code 0
Press ENTER to exit console.
```



Scanned with OKEN Scanner